

Laplace Transformation

Definition:

Let $f(t)$ be function defined for all positive values of t , then

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

provided the integral exists, is called the laplace transform of $f(t)$. It is denoted as

$$\mathcal{L}[f(t)] = F(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

Important formulae:

1. $\mathcal{L}(1) = \frac{1}{s}$
2. $\mathcal{L}(t^n) = \frac{n!}{s^{n+1}}$, when $n = 0, 1, 2, \dots$
3. $\mathcal{L}(e^{at}) = \frac{1}{s-a}$
4. $\mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}$
5. $\mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$
6. $\mathcal{L}(e^{at} t^n) = \frac{n!}{(s-a)^{n+1}}$

$$7. \mathcal{L}(e^{at} \sin bt) = \frac{b}{(s-a)^2 + b^2}$$

$$8. \mathcal{L}(e^{at} \cos bt) = \frac{s-a}{(s-a)^2 + b^2}$$

Example of different formulae:

Formula #2: $\mathcal{L}(t^n) = \frac{n!}{s^{n+1}}$

(a) $\mathcal{L}(t^2) = \frac{2!}{s^{2+1}} = \frac{2}{s^3}$

(b) $\mathcal{L}(t^4) = \frac{4!}{s^{4+1}} = \frac{24}{s^5}$

Formula #3: $\mathcal{L}(e^{at}) = \frac{1}{s-a}$

(a) $\mathcal{L}(e^t) = \frac{1}{s-1}$

(b) $\mathcal{L}(e^{2t}) = \frac{1}{s-2}$

(c) $\mathcal{L}(e^{-3t}) = \frac{1}{s-(-3)} = \frac{1}{s+3}$

(d) $\mathcal{L}(e^{\frac{t}{2}}) = \frac{1}{s-\frac{1}{2}}$

(e) $\mathcal{L}(e^{-\frac{t}{2}}) = \frac{1}{s-(-\frac{1}{2})} = \frac{1}{s+\frac{1}{2}}$

Formula # 4: $\mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}$

$$(a) \mathcal{L}(\sin 2t) = \frac{2}{s^2 + 2^2} = \frac{2}{s^2 + 4}$$

$$(b) \mathcal{L}(\sin 3t) = \frac{3}{s^2 + 3^2} = \frac{3}{s^2 + 9}$$

$$(c) \mathcal{L}\left(\sin \frac{t}{3}\right) = \frac{\frac{1}{3}}{s^2 + \left(\frac{1}{3}\right)^2} = \frac{\frac{1}{3}}{s^2 + \frac{1}{9}}$$

Formula # 5: $\mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$

$$(a) \mathcal{L}(\cos 2t) = \frac{s}{s^2 + 2^2} = \frac{s}{s^2 + 4}$$

$$(b) \mathcal{L}(\cos 3t) = \frac{s}{s^2 + 3^2} = \frac{s}{s^2 + 9}$$

$$(c) \mathcal{L}\left(\cos \frac{t}{2}\right) = \frac{s}{s^2 + \left(\frac{1}{2}\right)^2} = \frac{s}{s^2 + \frac{1}{4}}$$

Formula # 6: $\mathcal{L}(e^{at} t^n) = \frac{n!}{(s-a)^{n+1}}$

$$(a) \mathcal{L}(e^{2t} t) = \frac{1!}{(s-2)^{1+1}} = \frac{1}{(s-2)^2}$$

$$(b) \mathcal{L}(e^{-3t} t^2) = \frac{2!}{\{s - (-3)\}^{2+1}} = \frac{2}{(s+3)^3}$$

$$(c) \mathcal{L}(e^{\frac{t}{2}} t^2) = \frac{2!}{(s - \frac{1}{2})^{2+1}} = \frac{2}{(s - \frac{1}{2})^3}$$

Formula # 7: $\mathcal{L}(e^{at} \sin bt) = \frac{b}{(s-a)^2 + b^2}$

(a) $\mathcal{L}(e^{2t} \sin 3t) = \frac{3}{(s-2)^2 + 3^2}$

(b) $\mathcal{L}(e^{-3t} \sin 2t) = \frac{2}{\{s - (-3)\}^2 + 2^2} = \frac{2}{(s+3)^2 + 4}$

(c) $\mathcal{L}(e^{\frac{1}{2}t} \sin 4t) = \frac{4}{(s-\frac{1}{2})^2 + 4^2}$

(d) $\mathcal{L}(e^{-\frac{1}{3}t} \sin 2t) = \frac{2}{\{s - (-\frac{1}{3})\}^2 + 2^2} = \frac{2}{(s+\frac{1}{3})^2 + 4}$

(e) $\mathcal{L}(e^{\frac{1}{2}t} \sin \frac{t}{3}) = \frac{\frac{1}{3}}{(s-\frac{1}{2})^2 + (\frac{1}{3})^2}$

Formula # 8: $\mathcal{L}(e^{at} \cos bt) = \frac{s-a}{(s-a)^2 + b^2}$

(a) $\mathcal{L}(e^{2t} \cos 3t) = \frac{s-2}{(s-2)^2 + 3^2}$

(b) $\mathcal{L}(e^{-3t} \cos 2t) = \frac{s - (-3)}{\{s - (-3)\}^2 + 2^2} = \frac{s+3}{(s+3)^2 + 4}$

(c) $\mathcal{L}(e^{\frac{1}{2}t} \cos 4t) = \frac{s - \frac{1}{2}}{(s-\frac{1}{2})^2 + 4^2}$

(d) $\mathcal{L}(e^{-\frac{1}{3}t} \cos 2t) = \frac{s - (-\frac{1}{3})}{\{s - (-\frac{1}{3})\}^2 + 2^2} = \frac{s + \frac{1}{3}}{(s+\frac{1}{3})^2 + 4}$

prob-1:

prove that the following properties:

(a) $\mathcal{L}(1) = \frac{1}{s}$ (b) $\mathcal{L}(e^{at}) = \frac{1}{s-a}$

(c) $\mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}$ (d) $\mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$

Solution:

(a) $\mathcal{L}(1) = \frac{1}{s}$

Let $f(t) = 1$

We know,

$$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

$$\therefore \mathcal{L}(1) = \int_0^{\infty} e^{-st} \cdot 1 dt$$

$$= \int_0^{\infty} e^{-st} dt$$

$$= \left[\frac{e^{-st}}{-s} \right]_0^{\infty} \quad \left[\because \int e^{mx} dx = \frac{e^{mx}}{m} \right]$$

$$= -\frac{1}{s} [e^{-st}]_0^{\infty}$$

$$= -\frac{1}{s} [e^{-s(\infty)} - e^{-s(0)}]$$

$$= -\frac{1}{s} (e^{-\infty} - e^0)$$

$$= -\frac{1}{s} (0 - 1) \quad \left[\because e^{-\infty} = 0 \text{ and } e^0 = 1 \right]$$

$$\therefore \mathcal{L}(1) = \frac{1}{s} \text{ (proved)}$$

$$(b) \mathcal{L}(e^{at}) = \frac{1}{s-a}$$

$$\text{Let } f(t) = e^{at}$$

we know,

$$\mathcal{L}[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

$$\therefore \mathcal{L}(e^{at}) = \int_0^{\infty} e^{-st} \cdot e^{at} dt$$

$$= \int_0^{\infty} e^{-(s-a)t} dt$$

$$= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} \quad \left[\because \int e^{mx} dx = \frac{e^{mx}}{m} \right]$$

$$= -\frac{1}{s-a} \left[e^{-(s-a)t} \right]_0^{\infty}$$

$$= -\frac{1}{s-a} \left[e^{-(s-a)(\infty)} - e^{-(s-a)(0)} \right]$$

$$= -\frac{1}{s-a} (e^{-\infty} - e^0)$$

$$= -\frac{1}{s-a} (0 - 1) \quad \left[\because e^{-\infty} = 0 \text{ and } e^0 = 1 \right]$$

$$\therefore \mathcal{L}(e^{at}) = \frac{1}{s-a} \text{ (proved)}$$

$$(c) \mathcal{L}(\sin at) = \frac{a}{s^2 + a^2}$$

$$\therefore \mathcal{L}(\sin at) = \mathcal{L}\left[\frac{e^{iat} - e^{-iat}}{2i}\right] \quad \left[\because \sin x = \frac{e^{ix} - e^{-ix}}{2i}\right]$$

$$= \frac{1}{2i} \mathcal{L}(e^{iat} - e^{-iat})$$

$$= \frac{1}{2i} [\mathcal{L}(e^{iat}) - \mathcal{L}(e^{-iat})]$$

$$= \frac{1}{2i} \left[\frac{1}{s - ia} - \frac{1}{s + ia} \right] \quad \left[\because \mathcal{L}(e^{at}) = \frac{1}{s - a}\right]$$

$$= \frac{1}{2i} \left[\frac{s + ia - (s - ia)}{(s + ia)(s - ia)} \right]$$

$$= \frac{1}{2i} \left[\frac{s + ia - s + ia}{s^2 - (ia)^2} \right]$$

$$= \frac{1}{2i} \left(\frac{2ia}{s^2 - i^2 a^2} \right)$$

$$= \frac{a}{s^2 - i^2 a^2}$$

$$= \frac{a}{s^2 - (-1)a^2} \quad \left[\because i^2 = -1\right]$$

$$\therefore \mathcal{L}(\sin at) = \frac{a}{s^2 + a^2} \text{ (proved)}$$

$$(d) \mathcal{L}(\cos at) = \frac{s}{s^2 + a^2}$$

$$\therefore \mathcal{L}(\cos at) = \mathcal{L}\left[\frac{e^{iat} + e^{-iat}}{2}\right] \left[\because \cos x = \frac{e^{ix} + e^{-ix}}{2} \right]$$

$$= \frac{1}{2} \mathcal{L}[e^{iat} + e^{-iat}]$$

$$= \frac{1}{2} [\mathcal{L}(e^{iat}) + \mathcal{L}(e^{-iat})]$$

$$= \frac{1}{2} \left[\frac{1}{s - ia} + \frac{1}{s + ia} \right] \left[\because \mathcal{L}(e^{at}) = \frac{1}{s - a} \right]$$

$$= \frac{1}{2} \left[\frac{s + ia + s - ia}{(s + ia)(s - ia)} \right]$$

$$= \frac{1}{2} \left[\frac{2s}{s^2 - (ia)^2} \right]$$

$$= \frac{s}{s^2 - i^2 a^2}$$

$$= \frac{s}{s^2 - (-1)a^2}$$

$$\therefore \mathcal{L}(\cos at) = \frac{s}{s^2 + a^2} \text{ (proved)}$$

prob-2:

Find the Laplace transform of:

- (a) $1 + \cos 2t$ (b) $t \sin at$ (c) $t^v \sin at$ (d) $t^v \cos at$
 (e) $t \cos at$.

Solution:

$$(a) \mathcal{L}(1 + \cos 2t) = \mathcal{L}(1) + \mathcal{L}(\cos 2t)$$

$$= \frac{1}{s} + \frac{s}{s^2 + 2^2}$$

$$= \frac{1}{s} + \frac{s}{s^2 + 4}$$

$$= \frac{s^2 + 4 + s^2}{s^2(s^2 + 4)}$$

$$= \frac{2s^2 + 4}{s^2(s^2 + 4)}$$

$$\therefore \mathcal{L}(1 + \cos 2t) = \frac{2(s^2 + 2)}{s^2(s^2 + 4)} \text{ (Ans.)}$$

$$(b) \mathcal{L}(t \sin at) = \mathcal{L}\left(t \cdot \frac{e^{iat} - e^{-iat}}{2i}\right) \left[\because \sin x = \frac{e^{ix} - e^{-ix}}{2i}\right]$$

$$= \frac{1}{2i} [\mathcal{L}(t e^{iat}) - \mathcal{L}(t e^{-iat})]$$

$$= \frac{1}{2i} [\mathcal{L}(e^{iat} \cdot t) - \mathcal{L}(e^{-iat} \cdot t)]$$

$$= \frac{1}{2i} \left[\frac{1!}{(s - ia)^{1+1}} - \frac{1!}{(s + ia)^{1+1}} \right] \left[\because \mathcal{L}(e^{at} t^n) = \frac{n!}{(s - a)^{n+1}}\right]$$

$$\begin{aligned}
\therefore \mathcal{L}(t \sin at) &= \frac{1}{2i} \left[\frac{1}{(s-ia)^v} - \frac{1}{(s+ia)^v} \right] \\
&= \frac{1}{2i} \left[\frac{(s+ia)^v - (s-ia)^v}{(s+ia)^v (s-ia)^v} \right] \\
&= \frac{1}{2i} \left[\frac{(s^v + 2 \cdot s \cdot ia + i^v a^v) - (s^v - 2 \cdot s \cdot ia + i^v a^v)}{(s+ia)^v (s-ia)^v} \right] \\
&= \frac{1}{2i} \left[\frac{s^v + 2ias + (-1)a^v - s^v + 2ias - (-1)a^v}{(s^v - (ia)^v)^v} \right] \\
&= \frac{1}{2i} \left[\frac{s^v + 2ias - a^v - s^v + 2ias + a^v}{(s^v - i^v a^v)^v} \right] \\
&= \frac{1}{2i} \cdot \frac{4ias}{\{s^v - (-1)a^v\}^v} \quad [\because i^v = -1]
\end{aligned}$$

$$\therefore \mathcal{L}(t \sin at) = \frac{2as}{(s^v + a^v)^v} \quad (\text{Ans:})$$

$$\begin{aligned}
(d) \mathcal{L}(t^v \cos at) &= \mathcal{L} \left[t^v \cdot \frac{e^{iat} + e^{-iat}}{2} \right] \quad [\because \cos x = \frac{e^{ix} + e^{-ix}}{2}] \\
&= \frac{1}{2} [\mathcal{L}(t^v e^{iat}) + \mathcal{L}(t^v e^{-iat})] \\
&= \frac{1}{2} [\mathcal{L}(e^{iat} t^v) + \mathcal{L}(e^{-iat} t^v)] \\
&= \frac{1}{2} \left[\frac{2!}{(s-ia)^{2+1}} + \frac{2!}{(s+ia)^{2+1}} \right] \quad [\because \mathcal{L}(e^{at} t^n) = \frac{n!}{(s-a)^{n+1}}]
\end{aligned}$$

$$\begin{aligned}
\therefore \mathcal{L}(t^{\vee} \cos at) &= \frac{1}{2} \left[\frac{2}{(s-ia)^3} + \frac{2}{(s+ia)^3} \right] \\
&= \frac{1}{2} \cdot 2 \left[\frac{1}{(s-ia)^3} + \frac{1}{(s+ia)^3} \right] \\
&= \frac{(s+ia)^3 + (s-ia)^3}{(s+ia)^3 (s-ia)^3} \\
&= \frac{s^3 + 3s^{\vee}ia + 3s \cdot i^{\vee}a^{\vee} + i^3a^3 + s^3 - 3s^{\vee}ia + 3s \cdot i^{\vee}a^{\vee} - i^3a^3}{\{(s+ia)(s-ia)\}^3} \\
&= \frac{s^3 + 3ias^{\vee} + 3s(-1)a^{\vee} + (-i)a^3 + s^3 - 3ias^{\vee} + 3s(-1)a^{\vee} - (-i)a^3}{\{s^{\vee} - (ia)^{\vee}\}^3} \\
&= \frac{s^3 - 3sa^{\vee} - ia^3 + s^3 - 3sa^{\vee} + ia^3}{(s^{\vee} - i^{\vee}a^{\vee})^3} \\
&= \frac{2s^3 - 6sa^{\vee}}{\{s^{\vee} - (-1)a^{\vee}\}^3} \quad \left[\because i^{\vee} = -1 \right] \\
&= \frac{2s^3 - 6sa^{\vee}}{(s^{\vee} + a^{\vee})^3} \\
\therefore \mathcal{L}(t^{\vee} \cos at) &= \frac{2s(s^{\vee} - 3a^{\vee})}{(s^{\vee} + a^{\vee})^3} \text{ (Ans:)}
\end{aligned}$$